

IV Thrombolysis in the Extended Time Window for Acute Ischemic Stroke

An Updated Systematic Review and Meta-Analysis

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Abstract

Background and Objectives

IV thrombolysis (IVT) is the standard treatment for acute ischemic stroke (AIS) within 4.5 hours of symptom onset, yet many patients present beyond this window. Recent randomized-controlled clinical trials (RCTs) and updated guidelines suggest potential benefit of IVT in selected patients treated in the extended time window, but the overall evidence remains evolving. We aimed to evaluate the efficacy and safety of IVT beyond 4.5 hours from last known well.

Methods

We systematically searched MEDLINE, Scopus, and ClinicalTrials.gov for RCTs and individual patient data meta-analyses of RCTs comparing IVT plus best medical therapy (BMT) with BMT alone in patients with AIS presenting more than 4.5 hours after last known well. The primary efficacy outcome was excellent functional outcome at 90 days (modified Rankin Scale [mRS] score 0–1). Secondary efficacy outcomes included good functional outcome (mRS 0–2) and reduced disability (≥ 1 -point shift across the mRS). The primary safety outcome was symptomatic intracranial hemorrhage (sICH); secondary safety outcomes were any ICH and all-cause mortality at 90 days. Prespecified subgroup analyses were performed according to thrombolytic agent, treatment window, imaging selection strategy, vascular territory, and administration of endovascular treatment. Trial sequential analysis was conducted for the primary outcome to evaluate the robustness of the cumulative evidence.

Results

Thirteen studies were included, comprising 2,456 patients in the IVT group (mean age = 68.2 years; 37% female, mean baseline NIH Stroke Scale [NIHSS] score = 11.5) and 2,411 patients in the control group (mean age = 68.6 years; 38% female, mean baseline NIHSS score = 11.0). IVT significantly improved excellent functional outcome (risk ratio [RR] 1.23; 95% CI 1.15–1.33; $I^2 = 0\%$; NNT = 13), good functional outcome (RR 1.15; 95% CI 1.05–1.25; $I^2 = 59\%$), and reduced disability (common odds ratio 1.26; 95% CI 1.13–1.39; $I^2 = 0\%$). IVT was associated with increased risk of sICH (RR 2.11; 95% CI 1.36–3.28; $I^2 = 0\%$; NNH = 75). The rates of any ICH and all-cause mortality were similar between the 2 groups. No significant subgroup differences were documented across different subgroups. Trial sequential analysis confirmed that the cumulative evidence for the primary outcome crossed the efficacy boundary, supporting a robust treatment effect.

Discussion

IVT in the extended time window was associated with improved functional outcomes without increased mortality, supporting its use in selected patients despite a higher rate of sICH.

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Glossary

AIS = acute ischemic stroke; **BMT** = best medical therapy; **CTP** = CT perfusion; **DWI-FLAIR** = diffusion-weighted MRI and fluid-attenuated inversion recovery; **EVT** = endovascular treatment; **GRADE** = Grading of Recommendations, Assessment, Development, and Evaluations; **IPDM** = individual patient data meta-analysis; **IVT** = IV thrombolysis; **MRP** = magnetic resonance perfusion; **mRS** = modified Rankin Scale; **NIHSS** = NIH Stroke Scale; **NNH** = number needed to harm; **NNT** = number needed to treat; **PI** = prediction interval; **RCT** = randomized-controlled clinical trial; **RR** = risk ratio; **sICH** = symptomatic intracranial hemorrhage; **TSA** = trial sequential analysis.

Trial Registration Information

The prespecified protocol of the present systematic review and meta-analysis has been registered in the International Prospective Register of Ongoing Systematic Reviews (registration ID: CRD420261330511).

Introduction

IV thrombolysis (IVT) represents the fundamental systemic reperfusion therapy for acute ischemic stroke (AIS) when administered within 4.5 hours from symptom onset or last known well. Timely reperfusion within this window improves functional outcomes and reduces long-term disability, forming the basis of current international guideline recommendations.¹⁻³ Despite its established efficacy, strict adherence to a time-based threshold excludes a substantial proportion of patients, particularly those presenting beyond 4.5 hours or awakening with stroke symptoms of unknown onset.⁴

The traditional 4.5-hour paradigm is increasingly challenged by advances in neuroimaging that enable tissue-based rather than purely time-based treatment selection. Diffusion-weighted MRI and fluid-attenuated inversion recovery (DWI-FLAIR) mismatch serves as a surrogate marker of stroke onset within the conventional 4.5-hour treatment window,⁵ whereas perfusion imaging techniques (CTP/MRP) identify patients with salvageable ischemic penumbra despite delayed presentation.⁶ These developments have prompted multiple randomized-controlled clinical trials (RCTs) evaluating IVT in patients with unknown onset and in extended time windows based on imaging selection.

Recent aggregated evidence suggests that IVT beyond 4.5 hours may confer meaningful functional benefit in carefully selected patients. In a recent systematic review and meta-analysis of RCTs including patients treated beyond the standard window, IVT in combination with best medical therapy (BMT) significantly increased the likelihood of excellent functional outcome (modified Rankin Scale [mRS] scores of 0–1) and good functional outcome (mRS scores of 0–2), as well as improved overall disability shift, without an increase in all-cause mortality.⁷ Although symptomatic intracranial hemorrhage (sICH) occurred more frequently in the IVT group, this excess risk did not translate into higher disability, underscoring a clinically relevant net functional benefit in selected patients. It is important that the pooled

data were derived from RCT evidence, with overall low risk of bias and consistent effects across major subgroups, reinforcing the robustness of the findings.

Reflecting this evolving evidence base, the 2026 American Heart Association/American Stroke Association guidelines state that IVT may be reasonable in selected patients presenting beyond 4.5 hours from last known well based on advanced neuroimaging.³ This includes patients with wake-up or unknown-onset stroke identified through imaging-based selection strategies, such as perfusion imaging or MRI mismatch techniques. These Class IIa recommendations underscore moderate-quality evidence and highlight the shift from strictly time-based eligibility criteria toward tissue-based decision-making in late-presenting stroke.

Nevertheless, since the evidence synthesis informing these recommendations, additional RCTs have been completed and publicly reported, further expanding the data set and introducing variability in thrombolytic agents, imaging paradigms, and inclusion criteria. The heterogeneity across studies and the continuous emergence of new trial data necessitate an updated, comprehensive assessment of the aggregate randomized evidence.

In this context, we conducted a systematic review and aggregate-data meta-analysis comparing IVT BMT vs BMT alone in adults with AIS presenting more than 4.5 hours after last known well. Our objective was to assess the efficacy and safety profile of IVT in the extended time window and to clarify the balance between functional benefit and hemorrhagic risk in the context of contemporary randomized evidence. We also conducted trial sequential analysis (TSA) to evaluate the robustness and sufficiency of the cumulative evidence.

Methods

Standard Protocol Approvals, Registrations, and Patient Consents

The study protocol was prospectively registered in the International Prospective Register of Ongoing Systematic

Reviews (CRD420261330511; registered March 2, 2026). The conduct and reporting of this systematic review and meta-analysis adhered to the Preferred Reporting Items for Systematic Reviews and Meta-analyses guidelines.⁸ Because this work synthesized data from previously published RCTs, institutional review board approval and patient consent were not required.

Data Sources, Searches, and Study Selection

We conducted a systematic literature search according to the patient, intervention, comparison, and outcome (PICO) model⁹ to identify available RCTs or individual patient-data meta-analyses (IPDMs) of RCTs including adult patients with AIS presenting beyond 4.5 hours of last known well (including wake-up strokes or strokes of unknown onset; P: population) receiving IVT together with BMT (I: intervention) vs BMT alone with or without placebo (C: control) and investigating the outcomes of interest as outlined below (O: outcome). The thrombolytic agents in the IVT group could include alteplase or tenecteplase. BMT was defined as guideline-directed standard care for AIS, excluding IVT or other investigational pharmacologic agents. Studies allowing endovascular treatment (EVT) in either study arm were not excluded.

We excluded observational cohort studies, noncontrolled studies, case series, and case reports. Trials directly comparing different thrombolytic agents without a nonthrombolysis control arm were not eligible. Editorials, narrative reviews, and commentaries were excluded.

The literature search was performed independently by 4 reviewers (L.P., N.M.P., A.T., and G. Tsivgoulis). We systematically searched MEDLINE, Scopus, and ClinicalTrials.gov from database inception through March 3, 2026, using combinations of keywords related to “stroke,” “intravenous thrombolysis,” and “extended time window.” The full-search strategy is provided in eMethods. No language restrictions were applied.

To ensure completeness, reference lists of eligible publications and relevant conference abstracts were manually reviewed. Titles and abstracts were screened independently by 4 reviewers (L.P., N.M.P., M.R., and M.I.S.), followed by full-text evaluation of potentially eligible studies. Discrepancies were resolved through discussion, with adjudication by a fifth reviewer (G. Tsivgoulis) when necessary.

Quality Control, Bias Assessment, and Data Extraction

Methodologic quality and risk of bias were assessed using the Cochrane Risk of Bias 2 tool for RCTs.¹⁰ Assessments were performed independently by 4 reviewers (L.P., N.M.P., A.T., and M.I.S.), with disagreements resolved by consensus involving a fifth author (G. Tsivgoulis).

Data were extracted independently by 4 investigators using a predefined structured form. Extracted variables included study characteristics (trial name, country, and recruitment period), intervention and comparator details, eligibility imaging criteria, sample size, and baseline participant characteristics (age, sex, baseline NIH Stroke Scale [NIHSS] score, and time from last known well). Outcome data were collected for all prespecified outcomes.

Outcomes

The primary efficacy outcome of interest was excellent functional outcome at 3 months, defined as mRS scores of 0–1.¹¹ Good functional outcome at 3 months (defined as mRS scores ≤ 2)¹¹ and reduced disability at 3 months (defined as at least 1-point reduction across all mRS strata)¹¹ were assessed as secondary efficacy outcomes.

The primary safety outcome was sICH, as defined in each individual trial. Secondary safety outcomes included any ICH and all-cause mortality at 90 days.

Statistical Analysis

We performed pairwise meta-analyses comparing IVT plus BMT with BMT alone. For dichotomous outcomes, pooled risk ratios (RRs) with 95% CIs were calculated. Event rates in each arm were summarized using variance-stabilizing double-arcsine transformation before pooling. For ordinal disability shift analysis, unadjusted common odds ratios (cORs) were estimated using generic inverse-variance methods. Random-effects models (DerSimonian-Laird) were applied to account for between-study heterogeneity.¹² Prediction intervals (PIs) were calculated for all outcomes of interest. Statistical heterogeneity was quantified using the I^2 statistic and Cochran Q test, with I^2 values of <25%, 25%–50%, and >50% interpreted as low, moderate, and substantial heterogeneity, respectively. A 2-sided p value <0.05 was considered statistically significant (Q test threshold $p < 0.10$).

Prespecified subgroup analyses were conducted according to thrombolytic agent (alteplase vs tenecteplase), time window definition (4.5–9 hours, 4.5–24 hours, and >4.5 hours with no upper limit and wake-up stroke), imaging selection strategy (advanced imaging, including CTP/MRP mismatch or DWI/FLAIR MRI mismatch vs standard imaging including non-contrast CT), affected circulation (anterior, posterior, or both), and EVT treatment. Notably, CTA, MRA, or digital subtraction angiography was not considered as advanced imaging modalities for the purpose of this classification because they were used to assess vascular occlusion status rather than to define tissue viability or estimate stroke onset timing. In addition, a post hoc subgroup analysis was performed to further stratify advanced imaging into perfusion-based imaging (CTP/MRP) and DWI/FLAIR mismatch, to explore potential differences between tissue-based and time-based imaging paradigms. A leave-one-out sensitivity analysis was also performed to assess the robustness of the pooled

estimates by sequentially excluding each study from the meta-analysis. All analyses were based on intention-to-treat data.

For the primary efficacy outcome, the number needed to treat (NNT) was calculated using the following formula: $NNT = \frac{1}{(RR - 1)} \times \text{rate in BMT group}$ as previously described.¹³ Similarly, for the primary safety outcome, the number needed to harm (NNH) was calculated using the following formula: $NNH = \frac{1}{(RR - 1)} \times \text{rate in BMT group}$.

Baseline comparability between study arms was evaluated using odds ratios for categorical variables and standardized mean differences for continuous variables. For studies reporting medians and interquartile ranges, means and SDs were estimated using established quantile-based methods.¹⁴

Small study effects were evaluated when at least 4 studies contributed to an outcome, using funnel plot inspection and Egger regression testing.¹⁵ All meta-analyses were performed using R (version 3.5.0; package meta).¹⁶

For each outcome of interest, the risk of bias was assessed and the quality of evidence was rated as high, moderate, low, or very low based on the type of available evidence and considerations on inconsistency of results, indirectness of evidence, imprecision of results, and risk of bias, according to the Grading of Recommendations, Assessment, Development, and Evaluations (GRADE) methodology.¹⁷

To assess the robustness of the primary efficacy findings and account for potential type I and type II errors due to cumulative testing, we performed TSA. This approach applies monitoring boundaries analogous to interim analyses in a single trial and estimates the required information size necessary to confirm or refute a predefined intervention effect.^{18,19} The following parameters were prespecified: (1) type I error risk, set to 5%; (2) type II error risk, set to 10%; (3) control event proportion, derived from the accumulated control groups of the included trials; (4) heterogeneity between trials, taken from the empirical distribution of heterogeneity variance estimates as estimated from Cochrane reviews²⁰; and (5) anticipated intervention effect, set as risk difference = 3%, as previously described,² and the corresponding RR in the proportions of patients in mRS 0–1 between the 2 treatment groups. Continuity corrections were applied for zero-event trials using empirical adjustment methods.²¹ Specifically, we added an adjustment factor of 0.5 to the number of events in the TNK group and in the TPA group in case of zero events in 1 group. The above statistical analyses were performed using R software version 3.5.0 (package: RTSA).²²

Data Availability

All data analyzed or generated in this study are included in this article and its supplementary material files.

Results

Literature Search and Included Studies

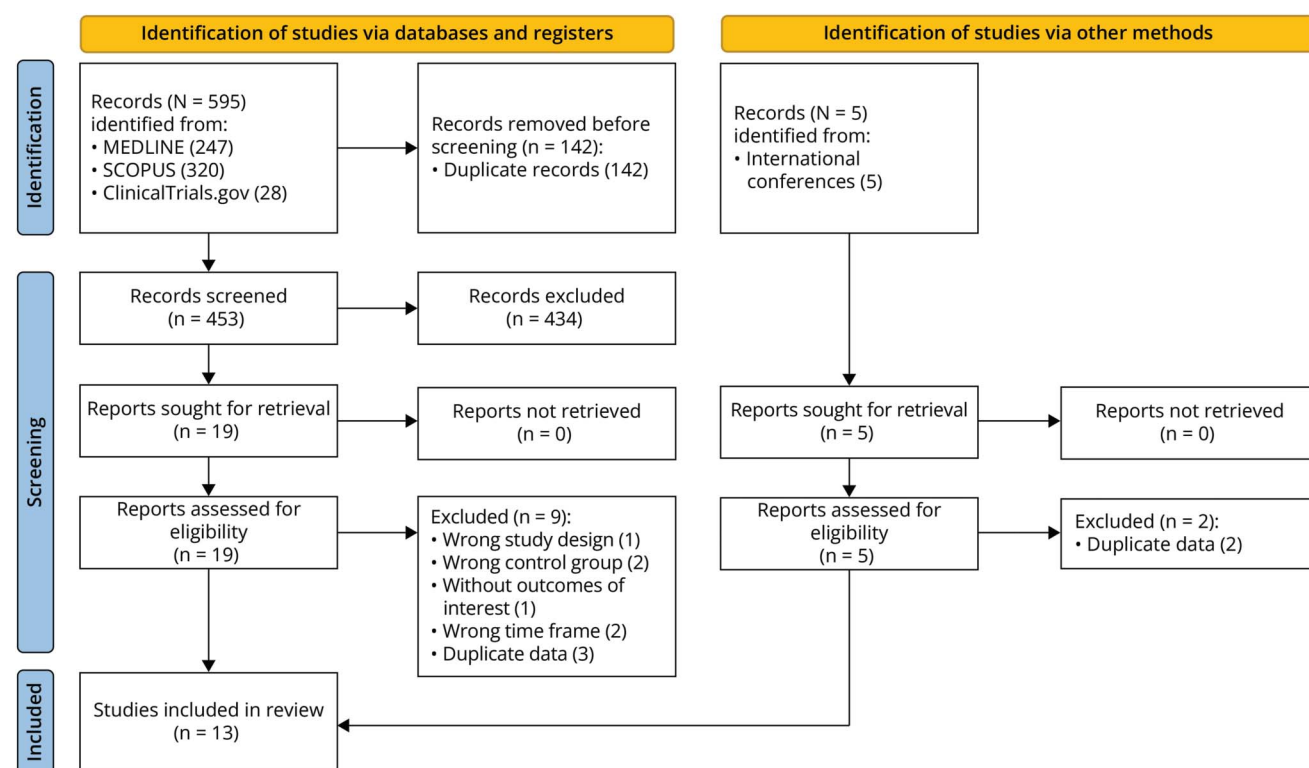
The systematic database search yielded a total of 247, 320, and 28 records from the MEDLINE SCOPUS and ClinicalTrials.gov databases, respectively (Figure 1). After excluding duplicates and initial screening, we retrieved the full text of 19 records that were considered potentially eligible for inclusion. After reading the full-text articles, 9 were further excluded (eTable 1). Furthermore, after searching the reference lists of published articles and international conference abstracts, 5 additional records were considered potentially eligible, with 3 of them finally included. Finally, we identified 12 eligible RCTs and 1 IPDM of RCTs (pooling data from 3 RCTs: Extending the Time for Thrombolysis in Emergency Neurological Deficits [EXTEND],²³ European Cooperative Acute Stroke Study-4: Extending the time for thrombolysis in emergency neurological deficits [ECASS4-EXTEND],²⁴ and Echoplanar Imaging Thrombolytic Evaluation Trial [EPITHET]²⁵) for inclusion in the systematic review and meta-analysis (Table 1).^{26–38} This systematic review and meta-analysis included a total of 4,867 patients with AIS who were presented beyond 4.5 hours from the time they were last known well, receiving either IVT together with BMT (n = 2,456; mean age 68.2 years; 37% female, mean baseline NIHSS score of 11.5; presenting 10.7 hours since they were last known well; EVT performed in 16%) vs BMT alone (n = 2,411; mean age 68.6 years; 38% female, mean baseline NIHSS score of 11.0; presenting 11.1 hours since they were last known well; EVT performed in 16%; eFigures 1–5). There were no significant baseline differences in the 2 groups.

However, variability was observed across the included studies about key inclusion criteria. Specifically, studies differed in the vascular territory included (anterior circulation,^{27,33–35} posterior circulation,^{26,28,36} or both^{23,29,31,32,37,38}), as well as in the imaging-based selection strategies applied. Patient selection was based on perfusion mismatch criteria^{27,29,30,33–35} or DWI/FLAIR mismatch^{31,32,38} in studies using advanced imaging, whereas other studies relied on standard neuroimaging with noncontrast CT.^{26,28,36,37} In addition, differences in core volume thresholds and definitions of mismatch were noted among trials using advanced imaging. These characteristics are summarized in detail in Table 1.

Quality Control of Included Studies

The risk of bias assessment for the included RCTs is summarized in eFigure 6. All studies implemented appropriate randomization process and were therefore judged to be at low risk of bias in the randomization domain. The 3 trials included in the IPDM²⁷ and the Tenecteplase in Stroke Patients Between 4.5 and 24 Hours (TIMELESS) trial³³ used placebo-controlled designs and showed no evidence of deviations from the intended interventions, resulting in a low risk of bias for this domain. Although the Efficacy and Safety of MRI-Based Thrombolysis in Wake-Up Stroke (WAKE-UP) trial also used a placebo-controlled design, 79 patients were excluded after

Figure 1 Flowchart of the Systematic Review



randomization.³⁸ The remaining trials reported minor deviations from the intended interventions, leading to some concerns in this domain. Regarding missing outcome data, Intravenous Tenecteplase Plus EVT Versus EVT Alone on 4.5 to 24 Hours After Basilar Artery Occlusion (ATTENTION LATE),²⁶ Extending the Time Window for Thrombolysis in Posterior Circulation Stroke Without Early CT Signs (EXPECTS),²⁸ Intravenous Thrombolysis With rhTNK-tPA for Acute Non-large Vessel Occlusion in Extended Time Window (OPTION),³⁰ Tenecteplase Reperfusion Therapy in Acute Ischaemic Cerebrovascular Events-5 (TRACE 5),³⁶ WAKE-UP,³⁸ and the IPDM²⁷ reported some degree of missing data. However, in all cases, the proportion of missing data was less than 10% of the randomized population; therefore, this domain was judged to be at low risk of bias. The remaining trials reported no missing outcome data. All studies were considered to be at low risk of bias in the measurement of outcomes because outcome assessors were blinded to treatment allocation. Finally, the risk of bias due to selective reporting was assessed as low across all included studies.

Quantitative Analyses

Regarding the primary efficacy outcome, patients receiving IVT had significantly higher rates of achieving an excellent functional outcome at 3 months compared with those receiving BMT alone (RR 1.23; 95% CI 1.15–1.33; PI 1.14–1.34; $p < 0.01$; 13 studies; $I^2 = 0\%$; p for Cochran $Q = 0.73$; Figure 2). The NNT was 13 (95% CI 9–21). There were no significant subgroup differences based on the thrombolytic

agent used (p for subgroup differences = 0.36), the time window applied (p for subgroup differences = 0.91), the neuroimaging method used (p for subgroup differences = 0.78), the affected territory (p for subgroup differences = 0.98), or the administration of EVT (p for subgroup differences = 0.26; eFigure 7).

Concerning secondary efficacy outcomes, the rates of good functional outcome were significantly higher for patients receiving IVT in addition to BMT compared with those receiving BMT alone (RR 1.15; 95% CI 1.05–1.25; PI 0.87–1.51; $p = 0.002$; 13 studies; $I^2 = 59\%$; p for Cochran $Q < 0.01$; eFigure 8) with no subgroup differences presented. Furthermore, patients receiving IVT had significantly higher odds of reduced disability at 3 months compared with those receiving BMT alone (common odds ratio 1.26; 95% CI 1.13–1.39; PI 1.12–1.41; $p < 0.001$; 13 studies; $I^2 = 0\%$; p for Cochran $Q = 0.69$; Figure 3), with no subgroup differences detected (eFigure 9).

Regarding the primary safety outcome, the rate of sICH was significantly higher among patients treated with IVT (RR 2.11; 95% CI 1.36–3.28; PI 1.29–3.46; $p < 0.001$; 13 studies; $I^2 = 0\%$; p for Cochran $Q = 0.60$; Figure 4). The NNH was 75 (95% CI 37–231). Subgroup analyses did not reveal significant differences based on the thrombolytic agent used (p for subgroup differences = 0.18), the time window applied (p for subgroup differences = 0.42), the neuroimaging method (p for subgroup differences = 0.11), the affected circulation (p

Table 1 Baseline Characteristics of Studies Included in This Systematic Review and Meta-Analysis

Study	Country	Recruitment period	Imaging	Inclusion	Drug	Group	Age, y, mean (SD) or median (IQR)	Female, n	NIHSS, median (IQR)	EVT, n	LKW, median (IQR)
ATTENTION LATE ²⁶	China	March 2023–February 2025	CT and CTA/MRA/DSA	Acute basilar artery occlusion planned to undergo EVT; onset 4.5–24 h; NIHSS ≥10; mRS <2; pcASPECTS ≥6 on CT/CTA or ≥5 on DWI-MRI	Tenecteplase 0.25 mg/kg (maximum dose 25 mg)	IVT (n = 166)	65 ± 11	39	21 (12–35)	166	458 (337–686) min
						BMT (n = 164)	67 ± 11	50	22 (15–35)	164	472 (336–769) min
Campbell et al. (IPDM) ²⁷											
ECASS4-EXTEND ²⁴	Europe	January 2014–September 2017	MRI	Anterior circulation AIS; onset 4.5–9 h; NIHSS 4–26; perfusion volume >20 mL and mismatch ≥20%	Alteplase 0.9 mg/kg (maximum dose 90 mg)	IVT (n = 213)	73.2 ± 12.2	94	12 (7–17)	—	471 (346–485) min
EPITHET ²⁵	Australia, New Zealand, Belgium, UK	April 2001–January 2007	MRI	Anterior circulation AIS; onset 3–6 h; NIHSS ≥4; mRS <3; perfusion volume >10 mL and mismatch ≥20%							
EXTEND ²³	Australia, New Zealand, Taiwan, Finland	August 2010–June 2018	CTP, MRI	Anterior circulation AIS; onset 4.5–9 h; NIHSS 4–26; mRS <2; perfusion volume >10 mL and mismatch ≥20% and ischemic core <70 mL		PCB (n = 201)	72 ± 12.3	85	10 (6–16)	—	413 (353–480) min
EXPECTS ²⁸	China	August 2022–May 2024	CT	Posterior circulation AIS; onset 4.5–24 h; NIHSS ≥0; mRS <2; PC-ASPECTS ≥6	Alteplase 0.9 mg/kg (maximum dose 90 mg)	IVT (n = 117)	64 (57–76)	42	3 (2–6)	—	NA
						BMT (n = 117)	63 (55–74)	39	3 (1–6)	—	NA
HOPE ²⁹	China	June 2021–June 2024	CTP	Anterior and posterior circulation AIS; onset 4.5–24 h; NIHSS 4–26; mRS <2; ischemic core volume <70 mL and penumbra volume >10 mL and mismatch ≥20%	Alteplase 0.9 mg/kg (maximum dose 90 mg)	IVT (n = 186)	72 (62–80)	84	10 (6–15)	—	411 (328–551) min
						BMT (n = 186)	73 (65–80)	76	10 (6–14)	—	427 (352–542) min
OPTION ³⁰	China	June 2023–August 2025	CTP	Anterior and posterior circulation AIS; onset 4.5–24 h; NIHSS 6–25 or 4–5 with disabling deficit; mRS <2; ischemic core volume <50 mL and ratio of ischemic tissue to the initial infarct volume >1.2 and penumbra volume >10 mL; LVO excluded	Tenecteplase 0.25 mg/kg (maximum dose 25 mg)	IVT (n = 282)	69 (59–75)	106	7 (5–9)	6	12.4 (8.7–16.8) h
						BMT (n = 284)	67 (59–74)	90	6 (5–9)	1	11.5 (8.3–16.6) h

Continued

Table 1 Baseline Characteristics of Studies Included in This Systematic Review and Meta-Analysis (continued)

Study	Country	Recruitment period	Imaging	Inclusion	Drug	Group	Age, y, mean (SD) or median (IQR)	Female, n	NIHSS, median (IQR)	EVT, n	LKW, median (IQR)
ROSE-TNK ³¹	China	March 2021–July 2022	MRI	Anterior and posterior circulation AIS; onset 4.5–24 h; NIHSS 6–25; mRS <2; infarct <1/3 of MCA or <1/2 of ACA or <1/2 of PCA and ischemic core volume <70 mL and DWI-FLAIR mismatch	Tenecteplase 0.25 mg/kg (maximum dose 25 mg)	IVT (n = 40)	62.7 ± 8.9	9	7.5 (6–10.75)	—	NA
						BMT (n = 40)	62.8 ± 5.6	14	7 (6–8.75)	—	NA
THAWS ³²	Japan	May 2014–July 2018	MRI	Anterior and posterior circulation AIS; >4.5 h from symptoms onset; NIHSS 2–24; DWI-FLAIR MRI mismatch	Alteplase 0.6 mg/kg	IVT (n = 70)	73.2 ± 12.4	25	N/A	—	NA
						BMT (n = 61)	75.8 ± 11.90	30	N/A	—	NA
TIMELESS ³³	Canada, the United States	March 2019–December 2022	CTP, MRI	Anterior circulation AIS; onset 4.5–24 h; NIHSS >5; mRS <3; occlusion of ICA, M1, M2 on CTA or MRA; ischemic core volume <70 mL and penumbra volume >15 mL and ratio of ischemic tissue to the initial infarct volume >1.8	Tenecteplase 0.25 mg/kg (maximum dose 25 mg)	IVT (n = 228)	72 (62–79)	122	12 (8–17)	176	12.7 (9.2–15.8) h
						PCB (n = 230)	73 (63–82)	123	12 (8–18)	178	13 (9–16.9) h
TNK PLUS ³⁴	China	January 2024–July 2025	CTP OR MRI + MRP	Anterior circulation AIS; onset 4.5–24 h; NIHSS 6–25; mRS <3; M1-MCA or proximal M2-MCA occlusion planned to receive EVT; ischemic core volume <70 mL and mismatch ratio >1.8 and mismatch volume >15 mL	Tenecteplase 0.25 mg/kg (maximum dose 25 mg)	IVT (n = 199)	68 (58–74)	77	14 (9–17)	188	9.3 (6.4–14.3) h
						BMT (n = 192)	68.5 (60–75)	78	13 (10–17)	185	10.8 (6.7–14.3) h
TRACE III ³⁵	China	January 2022–November 2023	CTP, MRI	Anterior circulation AIS; onset 4.5–24 h; NIHSS 6–25; mRS <2; occlusion of ICA, M1, M2 on CTA or MRA; perfusion mismatch on CTP or DWI-MRI and ischemic core volume <70 mL and ratio of hypoperfused tissue to ischemic core volume >1.8 and penumbra volume >15 mL	Tenecteplase 0.25 mg/kg (maximum dose 25 mg)	IVT (n = 264)	67 (58–75)	81	11 (7–15)	—	12.4 (8.8–16.3) h
						BMT (n = 252)	68 (59–76)	85	10 (7–14)	—	12.8 (9–17.5) h

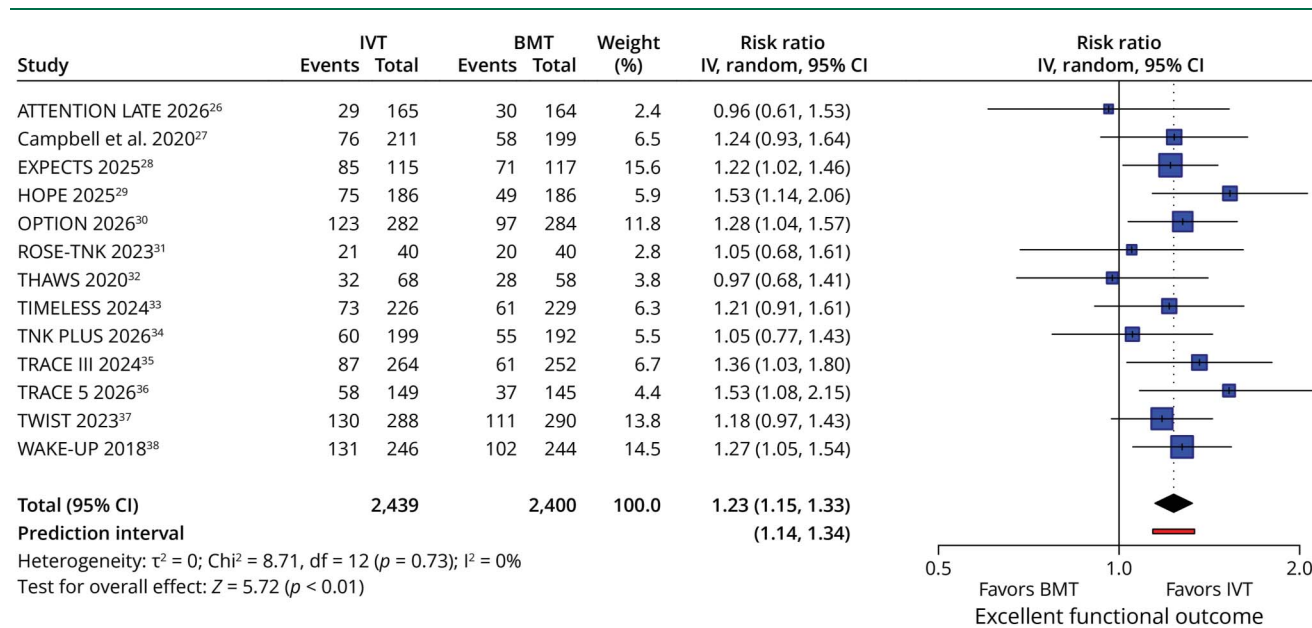
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Table 1 Baseline Characteristics of Studies Included in This Systematic Review and Meta-Analysis (*continued*)

Study	Country	Recruitment period	Imaging	Inclusion	Drug	Group	Age, y, mean (SD) or median (IQR)	Female, n	NIHSS, median (IQR)	EVT, n	LKW, median (IQR)
TRACE 5 ³⁶	China	January 2024–June 2025	CT, CTA or MRA	Basilar artery occlusion on CTA or MRA; onset 0–24 h; mRS <4; pc-ASPECTS >5	Tenecteplase 0.25 mg/kg (maximum dose 25 mg)	IVT (n = 149)	68.1 (60.2–74.2)	47	11 (6–23)	69	9.3 (6–13.8) h
						BMT (n = 145)	67.3 (60.4–74.2)	39	10 (6–18)	64	9.8 (7.3–13.6) h
TWIST ³⁷	International	June 2017–September 2021	CT	Anterior and posterior circulation AIS; onset on awakening; NIHSS >2 or aphasia	Tenecteplase 0.25 mg/kg (maximum dose 25 mg)	IVT (n = 288)	72.7 ± 11.3	124	6 (5–11)	18	NA
						BMT (n = 290)	72.9 ± 11.6	122	6 (5–10)	42	NA
WAKE-UP ³⁸	Europe	September 2012–June 2017	MRI	Anterior and posterior circulation AIS; onset on awakening or unknown onset but >4.5 h of last known well; NIHSS <26; mRS <2; DWI-FLAIR mismatch	Alteplase 0.9 mg/kg (maximum dose 90 mg)	IVT (n = 254)	65.3 ± 11.2	89	6 (4–9)	—	10.3 (8.1–12) h
						PCB (n = 249)	65.2 ± 11.9	89	6 (4–9)	—	10.4 (8.1–12.1) h

Abbreviations: ACA = anterior cerebral artery; AIS = achieved information size; ASPECTS = Alberta Stroke Program Early CT Score; ATTENTION LATE = Intravenous Tenecteplase Plus EVT Versus EVT Alone on 4.5 to 24 Hours After Basilar Artery Occlusion; BMT = bestmedical treatment; CTA = computed tomography angiography; CTP = CT perfusion; DSA = digital subtraction angiography; DWIFLAIIR = diffusion-weighted MRI and fluid-attenuated inversion recovery; ECASS4-EXTEND = European Cooperative Acute Stroke Study-4: Extending the time for thrombolysis in emergency neurological deficits; EPITHET = Echoplanar Imaging Thrombolytic Evaluation Trial; EVT = endovascular treatment; EXPECTS = EXtending the TimeWindow for Thrombolysis in Posterior Circulation Stroke Without Early CT Signs; EXTEND = Extending the Time for Thrombolysis in Emergency Neurological Deficits; HOPE = 24 Hours Treatment with Alteplase in Patients with Ischemic Stroke; ICA = internal carotid artery; IPDM = individual patient-data meta-analysis; IQR = interquartile range; IVT = IV thrombolysis; LKW = last-known-well; LVO = large vessel occlusion; M1 = M1 segment of the middle cerebral artery; M2 = M2 segment of the middle cerebral artery; MCA = middle cerebral artery; MRA = magnetic resonance angiography; mRS = modified Rankin Scale; NA = not applicable; NIHSS = NIH Stroke Scale; OPTION = Intravenous Thrombolysis With rhTNK-tPA for Acute Non-large Vessel Occlusion in Extended Time Window; OR = odds ratio; PCA = posterior circulation artery; pc-ASPECTS = posterior circulation Alberta Stroke Program Early CT Score; PCB = placebo; ROSETNK = MRI-guided thrOmbolysis for Stroke bEeyond Time Window by TNK; SE = standard error; SW = stage wise; THAWS = THrombolysis for Acute Wake-up and Unclear-onset Strokes With Alteplase at 0.6 mg/kg Trial; TIMELESS = Tenecteplase in Stroke Patients Between 4.5 and 24 Hours; TNK PLUS = Endovascular Treatment With or Without Preceding Intravenous Tenecteplase (TNK) in PatientsWith Late-window acUte Ischemic Stroke Due to Middle Cerebral Artery Occlusion; TRACE 5 = Tenecteplase Reperfusion Therapy in Acute Ischaemic Cerebrovascular Events-5; TRACE III = Tenecteplase Reperfusion Therapy in Acute Ischemic Cerebrovascular Events-III; TWIST = Tenecteplase in Wake-up Ischaemic Stroke Trial; WAKE-UP = Efficacy and Safety of MRI-Based Thrombolysis in Wake-Up Stroke.

Figure 2 Forest Plot Presenting the Risk Ratio of Excellent Functional Outcome at 3 Months Among Patients Receiving IVT Plus BMT vs BMT Alone

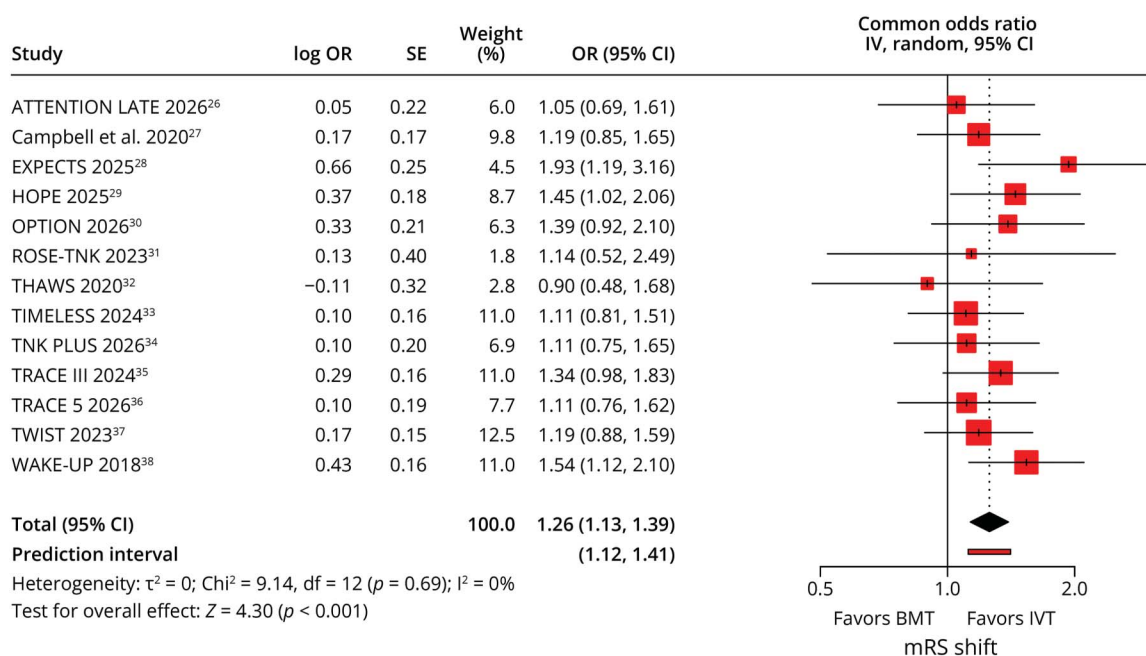


BMT = best medical treatment; IVT = IV thrombolysis.

for subgroup differences = 0.17), or the definition used (p for subgroup differences = 0.31), with the exception of significant subgroup differences when stratified for EVT (p for subgroup differences = 0.02; eFigure 10).

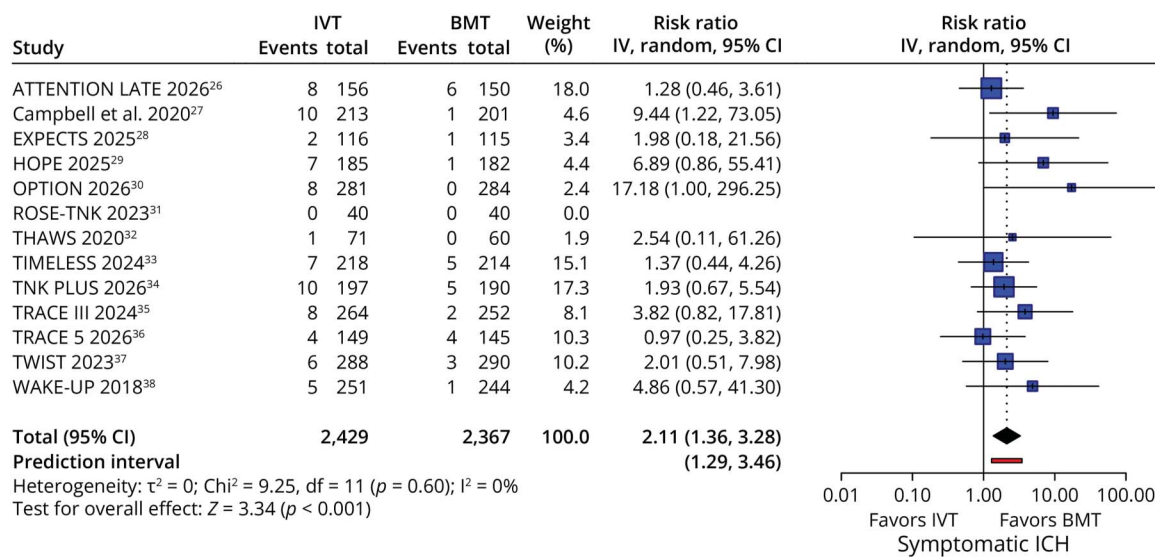
For secondary safety outcomes, the rates of any ICH did not significantly differ between groups (RR 1.32; 95% CI 1.00–1.74; PI 0.84–2.07; $p = 0.051$; 4 studies; $I^2 = 0\%$; p for Cochran $Q = 0.55$; eFigure 11), with no subgroup differences

Figure 3 Forest Plot Presenting the Common Odds Ratio of Reduced Disability at 3 Months Among Patients Receiving IVT Plus BMT vs BMT Alone



BMT = best medical treatment; IVT = IV thrombolysis; mRS = modified Rankin Scale; OR = odds ratio; SE = standard error.

Figure 4 Forest Plot Presenting the Risk Ratio of Symptomatic ICH Among Patients Receiving IVT Plus BMT vs BMT Alone



BMT = best medical treatment; ICH = intracranial hemorrhage; IVT = IV thrombolysis.

presented. In addition, all-cause mortality at 3 months was similar between the 2 groups (RR 1.03; 95% CI 0.89–1.18; PI 0.88–1.20; $p = 0.72$; 13 studies; $I^2 = 0\%$; p for Cochran $Q = 0.63$; eFigure 12).

To further explore potential differences within advanced imaging strategies, we performed a post hoc subgroup analysis stratifying studies into standard imaging, perfusion-based imaging, and DWI/FLAIR mismatch. No significant subgroup differences were observed across all outcomes (eFigure 13), with consistent direction of treatment effect across all imaging paradigms.

To assess the robustness of the findings, we performed leave-one-out sensitivity analyses for all efficacy and safety outcomes (eFigure 14). For the primary efficacy outcome, as well as for good functional outcome, reduced disability, sICH, and all-cause mortality, the direction and statistical significance of the pooled estimates remained unchanged after sequential omission of each individual study, indicating that no single trial disproportionately influenced the results. By contrast, for any ICH, the pooled estimate was borderline nonsignificant in the primary analysis, and the results of the leave-one-out analysis showed that omission of individual studies led to fluctuations around the threshold of statistical significance.

An overview of the analyses for the primary and secondary outcomes, with the respective GRADE assessments, is presented in Table 2. The pooled proportions per arm for each outcome of interest are presented in eTable2.

Assessment of publication bias using funnel plot inspection revealed no evidence of asymmetry for any of the outcomes, except for the outcome of good functional outcome at 90 days

(eFigures 15–20). For this outcome, visual asymmetry was observed due to the EXPECTS²⁸ study, without evidence of small study effects (p for the Egger test = 0.97). On the other hand, small study effects appeared in the evaluation of the outcome of sICH (p for the Egger test = 0.01).

Finally, TSA demonstrated that both the conventional boundaries ($Z = 1.96$) and the sequential monitoring boundaries were surpassed (Figure 5). This analysis demonstrates that the meta-analysis is conclusive as to whether the intervention (i.e., IVT in the extended time window) is statistically significant considering a realistic intervention effect, sparse data, and multiple testing.

Discussion

In this systematic review and meta-analysis including 12 RCTs and 1 IPDM of RCTs comprising 4,867 patients with AIS presenting beyond the conventional 4.5-hour time window from last known well, IVT was significantly associated with improved functional outcomes. Specifically, IVT increased the rates of excellent functional outcome at 90 days (RR 1.23; 95% CI 1.15–1.33; NNT = 13), as well as good functional outcome and overall reduction in disability across the mRS spectrum. Although IVT was associated with a higher risk of sICH (RR 2.11; 95% CI 1.36–3.28; NNH = 75), it was not associated with a statistically significant increase in the risk of any ICH or all-cause mortality at 3 months. Notably, there was no evidence of heterogeneity ($I^2 = 0\%$) across the primary efficacy and secondary efficacy or safety outcomes with the sole exception of good functional outcome. Taken together, these findings support a favorable benefit-risk profile for IVT in selected patients treated beyond the traditional 4.5-hour treatment window.

The observed improvement in functional outcomes suggests that the therapeutic window for thrombolysis may extend beyond rigid time thresholds when appropriate patient selection strategies are applied. In addition to the primary endpoint of excellent functional outcome, IVT was associated with higher rates of good functional outcome and a favorable shift across the entire mRS distribution, indicating a consistent treatment effect across multiple measures of functional recovery. These concordant efficacy signals strengthen the robustness of the findings.

Subgroup analyses demonstrated consistent treatment effects across thrombolytic agents, with similar benefits observed for tenecteplase (0.25 mg/kg) and alteplase (0.9 mg/kg). Similarly, there was no significant difference between imaging selection strategies, with IVT associated with improved functional outcomes in both standard and advanced neuroimaging settings (eFigure 7, panel C). However, interpretation of these findings requires caution. Studies using standard neuroimaging did not include unselected patients but applied clinical and CT-based criteria to exclude hemorrhage and large established infarction. Three of the 4 trials conducted using standard neuroimaging predominantly included patients with posterior circulation stroke, whereas most trials enrolling patients with anterior circulation stroke relied on advanced imaging selection. Although no statistically significant subgroup difference between anterior and posterior circulation was detected, these differences in trial design suggest that advanced imaging remains the most appropriate strategy for selecting patients with anterior circulation stroke until further randomized evidence becomes available.

Within advanced imaging strategies, perfusion-based imaging and DWI/FLAIR mismatch represent distinct biological paradigms, the former identifying potentially salvageable tissue and the latter serving as a surrogate marker of stroke onset timing. In post hoc analyses further stratifying these approaches, no significant subgroup differences were observed across efficacy and safety outcomes, with consistent direction of treatment effect irrespective of the imaging modality used for patient selection. These findings suggest that the benefit of IVT in the extended time window may be preserved across different imaging-based selection strategies, supporting the generalizability of our results in settings where either tissue-based or time-based imaging paradigms are applied, although these findings should be interpreted in the context of limited power for subgroup comparisons.

Another important source of heterogeneity among the included trials was the allowance of EVT. To better address the potential confounding effect of concomitant EVT treatment, we performed additional subgroup analyses stratified according to EVT use, based on studies with extractable outcome data. Across all efficacy outcomes, including excellent and good functional outcome as well as reduced disability, no significant subgroup differences were observed, with consistent direction of treatment effect irrespective of

EVT status. Overall, these findings suggest that the beneficial effect of IVT in the extended time window is preserved regardless of concomitant EVT use, supporting the robustness and generalizability of our results in contemporary reperfusion practice.

To further evaluate the stability of our findings, we performed TSA for the primary efficacy outcome. The cumulative evidence crossed the efficacy boundary, indicating that the available randomized data are sufficient to support a beneficial effect of IVT on excellent functional outcome in the extended time window. These results suggest that the observed benefit is unlikely to represent a spurious finding due to random error or repeated testing in cumulative meta-analysis. Nevertheless, additional RCTs remain important to refine patient selection criteria and optimize treatment strategies.

Several ongoing RCTs are expected to provide additional insights into the role of IVT in extended treatment windows and specific stroke subgroups. The Extending the Time Window for Intravenous Tenecteplase in Patients With Distal Medium Vessel Occlusions Stroke (TNK-MeVO) trial (NCT06559436) is evaluating tenecteplase vs BMT in patients with distal medium-vessel occlusion, a population that remains underrepresented in existing studies. Additional trials, including Extending the Time Window for Tenecteplase by Recanalization of Basilar Artery Occlusion in Posterior Circulation Stroke (POST-ETERNAL; NCT05105633), A Study to Test if Tenecteplase Helps People to Recover From an Acute Stroke When Given More Than 4.5 Hours After the Person Was Last Seen Well (TENACITY; NCT07361302), and Randomization to Extend Stroke Intravenous Thrombolysis In Evolving Non-Large Vessel Occlusion With TNK (RESILIENT-TNK; NCT05199662), are investigating the role of tenecteplase in extended time windows. These studies may further clarify optimal patient selection strategies, thrombolytic choice, and the influence of time from last known well on treatment efficacy and safety.

Regarding safety outcomes, IVT administered beyond 4.5 hours was associated with a higher risk of sICH, consistent with the known hemorrhagic risks of thrombolytic therapy. The pooled rate of sICH in the IVT arm was approximately 3%, which is comparable with rates reported in studies evaluating IVT within the conventional time window.³⁹⁻⁴¹ It is important that the increased risk of sICH did not translate into higher mortality at 3 months. When considered together with the NNT of 13 for excellent functional outcome and the NNH of 75 for sICH, these findings suggest an overall favorable clinical balance.

It is important that in the subgroup analyses, there were no significant differences in thrombolytic agent, treatment timing, neuroimaging modality, affected circulation, or definition of sICH. However, in the subgroup of studies using standard neuroimaging, there was no statistically significant increase in

Table 2 Overview of Analyses and GRADE Evidence Profile for Each Outcome of Interest

Certainty assessment							No. of patients		Effect		
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Intervention	Comparator	Relative (95% CI)	Absolute (95% CI)	Certainty
Excellent functional outcome (mRS scores 0–1 at 90 d): RR, unadjusted analysis											
13	12 RCTs and 1 IPDM of RCTs	Not serious	Serious ^a	Not serious	Not serious	None	980/2,439 (40.2%)	780/2,400 (32.5%)	RR 1.23 (1.15–1.33)	75 more per 1,000 (from 49 more to 107 more)	⊕⊕⊕○ Moderate
Good functional outcome (mRS scores 0–2 at 90 d): RR, unadjusted analysis											
13	12 RCTs and 1 IPDM of RCTs	Not serious	Serious ^a	Not serious	Not serious	None	1,334/2,439 (54.7%)	1,151/2,400 (48.0%)	RR 1.15 (1.05–1.25)	72 more per 1,000 (from 24 more to 120 more)	⊕⊕⊕○ Moderate
Reduced disability (≥1-point reduction across all mRS scores at 90 d): common OR, unadjusted analysis											
13	12 RCTs and 1 IPDM of RCTs	Not serious	Serious ^a	Not serious	Not serious	None	Not applicable	Not applicable	cOR 1.26 (0.13–1.39)	Not applicable	⊕⊕⊕○ Moderate
Symptomatic intracranial hemorrhage: RR, unadjusted analysis											
13	12 RCTs and 1 IPDM of RCTs	Not serious	Serious ^a	Not serious	Not serious	None	76/2,429 (3.1%)	29/2,367 (1.2%)	RR 2.11 (1.36–3.28) ^b	13 more per 1,000 (from 4 more to 27 more)	⊕⊕⊕⊕ High
Any intracranial hemorrhage: RR, unadjusted analysis											
4	4 RCTs	Not serious	Serious ^a	Not serious	Serious ^c	None	105/841 (12.5%)	79/839 (9.4%)	RR 1.32 (1.00–1.74)	30 more per 1,000 (from 0 more to 70 more)	⊕⊕○○ Low
All-cause mortality at 90 d: RR, unadjusted analysis											
13	12 RCTs and 1 IPDM of RCTs	Not serious	Serious ^a	Not serious	Serious ^c	None	321/2,438 (13.2%)	300/2,387 (12.6%)	RR 1.03 (0.89–1.18)	4 more per 1,000 (from 14 fewer to 23 more)	⊕⊕○○ Low

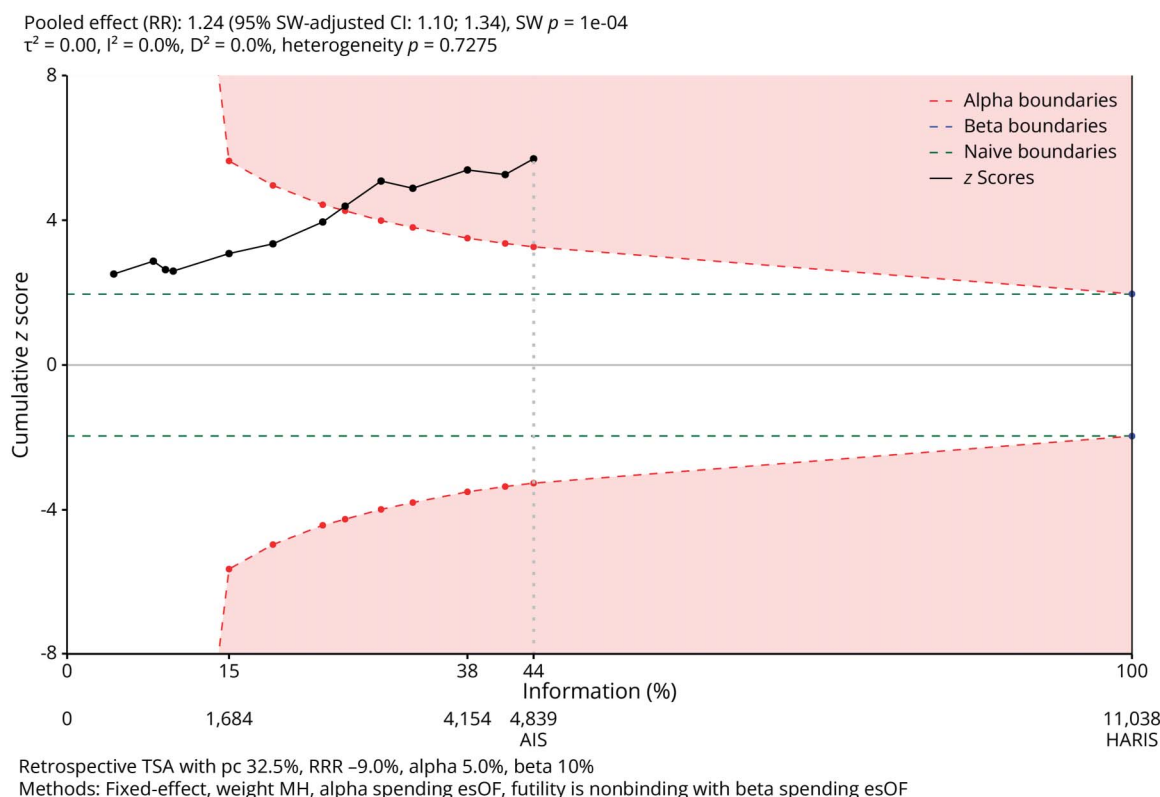
Abbreviations: cOR = common odds ratio; GRADE = Grading of Recommendations, Assessment, Development, and Evaluations; IPDM = individual patient-data meta-analysis; RCT = randomized-controlled clinical trial; RR = risk ratio.

^a Heterogeneous study designs.

^b Upgraded by 1 grade due to the strong association (RR > 2).

^c CIs including 1.

Figure 5 Trial Sequential Analysis for mRS 0–1 at 3 Months, Using Required Information Size



The red inward sloping lines to the left of the graph make up the trial sequential monitoring (O'Brien-Fleming) boundaries. The blue outward sloping lines with the corresponding blue area to the right of the graph represent the futility region. The z-statistic (summary estimate over its SE) is computed when a trial enters the analysis. The black line is the cumulative Z-curve, and each black square in the line represents 1 study. The number (in black) that is parallel to the end of the Z-curve represents the total number of patients in the corresponding meta-analysis. The intersection of the red lines at Z score +1.96 and –1.96 represents the "conventional" significant p value (0.05). AIS = achieved information size; esOF = Lan & DeMets version of O'Brien-Fleming; HARIS = heterogeneity adjusted required information size; MH = Mantel-Haenszel; mRS = modified Rankin Scale; OR = odds ratio; pc = control event proportion; RR = risk ratio; RRR = relative risk reduction; SE = standard error; TSA = trial sequential analysis.

the rates of sICH (RR 1.38; 95% CI 0.70–2.73; eFigure 10, panel C). This finding may again be explained by the inclusion of studies that were predominantly focused on posterior circulation strokes, which is further supported by the non-significant difference in sICH rates observed in the subgroup of posterior circulation strokes (RR 1.23; 95% CI 0.56–2.68; eFigure 10, panel D).

The relative resistance of posterior circulation strokes to sICH following IVT has previously been reported, with sICH occurring in 0.6% of posterior circulation strokes compared with 1.9% in anterior circulation strokes.⁴² Several mechanisms have been proposed to explain this observation.⁴³ Infarcts in the posterior circulation are often smaller and less frequently involve large cortical territories, which are known to be associated with a higher susceptibility to reperfusion injury after recanalization. In addition, posterior circulation ischemia more commonly involves brainstem or small perforator territories, including lacunar infarcts, which tend to produce smaller ischemic cores. These anatomical and pathophysiologic differences may collectively contribute to the lower rates of sICH reported in observational studies and meta-analyses comparing IVT outcomes between posterior and anterior circulation strokes.^{42,43}

In the additional subgroup analyses stratified by EVT use, a statistically significant interaction was observed for sICH (eFigure 10, panel F), with no apparent increase in sICH risk in studies where EVT was performed (RR 1.52; 95% CI 0.75–3.08). This finding should be interpreted with caution. In EVT-based trials, EVT was administered in both treatment arms, thereby attenuating differences in sICH risk that could be attributed specifically to IVT. By contrast, in studies without EVT, the increased risk of sICH more directly reflects the pharmacologic effect of IVT vs BMT (RR 5.03; 95% CI 2.35–10.77). Therefore, the observed interaction likely reflects differences in treatment context rather than a true modification of IVT-associated sICH risk by EVT. This finding is in line with observational and randomized data showing that bridging therapy (IVT and EVT) is not associated with increased risk of sICH compared with EVT alone.^{44,45}

Our findings are consistent with previous meta-analyses evaluating IVT in extended time windows.^{7,46,47} However, the present analysis provides several important advances. First, we incorporated all recently available RCTs, including studies presented at the 2026 International Stroke Conference that were not included in earlier evidence syntheses. Second, we

performed an extensive set of prespecified subgroup analyses exploring treatment effects across thrombolytic agents, imaging selection strategies, treatment windows, vascular territories, and EVT treatment. Third, the inclusion of TSA provides additional methodologic rigor by assessing the sufficiency and stability of the cumulative evidence.

An additional strength of our study is the formal evaluation of the certainty of evidence using the GRADE framework. Based on this assessment, the overall quality of evidence supporting the primary efficacy outcome and the secondary functional outcomes was rated as moderate, reflecting some heterogeneity in trial design and imaging selection strategies but consistent treatment effects across studies. The evidence regarding sICH was also considered robust, whereas the certainty for any ICH and all-cause mortality was lower due to imprecision in the pooled estimates. Overall, the GRADE assessment reinforces the consistency of the observed functional benefit associated with IVT in patients treated beyond the conventional 4.5-hour time window without evidence of increased mortality.

Despite its strengths, several limitations should also be acknowledged. This analysis was based on aggregated study-level data, and the included trials varied in inclusion criteria, imaging protocols, and outcome definitions. Although we conducted multiple subgroup analyses to explore potential sources of heterogeneity, such analyses remain limited by the lack of individual patient-level data. Future IPDM will be critical for refining patient selection and identifying subgroups most likely to benefit from IVT in extended time windows.

In conclusion, this comprehensive meta-analysis of randomized evidence suggests that IVT administered beyond the conventional 4.5-hour time window is associated with improved functional outcomes in selected patients with AIS, without an increase in mortality despite a higher risk of sICH. These findings support the extension of thrombolytic therapy beyond rigid time thresholds when appropriate patient selection strategies are applied.

Author Contributions

L. Palaiodimou: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data; investigation; methodology; visualization. N.M. Papageorgiou: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data; investigation; methodology; visualization. M. Romoli: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; investigation; methodology. A. Theodorou: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; investigation; methodology. M.I. Stefanou: drafting/revision of the manuscript for content, including medical writing for content; major role in

the acquisition of data; investigation; methodology. R. Lemmens: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; investigation. A. Sarraj: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; investigation. A.V. Alexandrov: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; investigation. G. Turc: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; investigation. U. Fischer: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; investigation. G. Tsivgoulis: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data; investigation, methodology, project administration, supervision.

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